

# Jihao Liu 刘济豪

## CURRICULUM VITAE

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## RESEARCH INTERESTS

Birational geometry, singularities, foliations, and explicit birational geometry.

## EMPLOYMENT

|                |                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------|
| 2024 – present | <b>Peking University</b> , School of Mathematical Sciences<br>Assistant Professor / Researcher.             |
| 2021 – 2024    | <b>Northwestern University</b> , Department of Mathematics<br>Boas Assistant Professor. Mentor: Yuchen Liu. |

## EDUCATION

|             |                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------|
| 2015 – 2021 | <b>University of Utah</b> , Salt Lake City, USA<br>Ph.D. in Mathematics. Advisor: Christopher D. Hacon FRS. |
| 2011 – 2015 | <b>Peking University</b> , Beijing, China<br>B.S. in Mathematics. Undergraduate advisor: Gang Tian.         |

## PUBLISHED PAPERS

Listed by year of journal publication, most recent first.

1. 2026 **Exceptional Fano varieties with small minimal log discrepancy.** Louis Esser, **Jihao Liu**, Chengxi Wang. *Math. Res. Lett.* **33** (2026), no. 1, 161–198. [arXiv:2406.03570](#), [doi:10.4310/MRL.260502161650](#).
2. 2026 **Sarkisov program for algebraically integrable and threefold foliations.** Yifei Chen, **Jihao Liu**, Yanze Wang. *Int. Math. Res. Not.* **2026**, no. 6, 1–19. [arXiv:2505.15115](#), [doi:10.1093/imrn/rnag045](#).
3. 2026 **A generalized non-vanishing theorem for surfaces.** **Jihao Liu**, Lingyao Xie. *Pure Appl. Math. Q.* (special volume in honor of Caucher Birkar) **22** (2026), no. 1, 235–244. [arXiv:2410.15457](#), [doi:10.4310/PAMQ.250306164159](#).
4. 2025 **Flop between algebraically integrable foliations on potentially klt varieties.** Yifei Chen, **Jihao Liu**, Yanze Wang. *Int. J. Math.* **36** (2025), 2550035. [arXiv:2410.05764](#), [doi:10.1142/S0129167X25500351](#).
5. 2025 **Minimal model program for algebraically integrable foliations on klt varieties.** **Jihao Liu**, Fanjun Meng, Lingyao Xie. *Compos. Math.* **161** (2025), 3213–3276. [arXiv:2404.01559](#), [doi:10.1112/S0010437X25102807](#).
6. 2025 **On the equivalence between the effective adjunction conjectures of Prokhorov–Shokurov and of Li.** Jingjun Han, **Jihao Liu**, Qingyuan Xue. *Algebra Number Theory* **19** (2025), no. 11. [arXiv:2312.15397](#), [doi:10.2140/ant.2025.19.2261](#).

7. 2025 **On explicit bounds of Fano threefolds.** Caucher Birkar, **Jihao Liu**. J. Reine Angew. Math. (Crelle's J.) (2025). [arXiv:2311.06732](#), [doi:10.1515/crelle-2025-0056](#).
8. 2025 **On termination of flips and exceptionally non-canonical singularities.** Jingjun Han, **Jihao Liu**. Geom. Topol. **29** (2025), no. 1, 399–441. [arXiv:2209.13122](#), [doi:10.2140/gt.2025.29.399](#).
9. 2025 **Uniform rational polytopes for Iitaka dimensions.** Guodu Chen, Jingjun Han, **Jihao Liu**. In: *Higher Dimensional Algebraic Geometry: A Volume in Honor of V. V. Shokurov* (C. D. Hacon, C. Xu eds.), London Math. Soc. Lecture Note Ser., Cambridge Univ. Press (2025), 43–68. [arXiv:2208.04663](#).
10. 2025 **Relative Nakayama–Zariski decomposition and minimal models for generalized pairs.** **Jihao Liu**, Lingyao Xie. Peking Math. J. (2025), no. 8, 299–349. [arXiv:2207.09576](#), [doi:10.1007/s42543-023-00076-2](#).
11. 2025 **On generalized lc pairs with b-log abundant nef part** (with an Appendix by J. Han). Junpeng Jiao, **Jihao Liu**, Lingyao Xie. Front. Math. (2025). [arXiv:2202.11256](#), [doi:10.1007/s11464-025-0033-0](#).
12. 2025 **ACC for minimal log discrepancies of terminal threefolds.** Jingjun Han, **Jihao Liu**, Yujie Luo. Adv. Math. **480** (2025), 110457. [arXiv:2202.05287](#), [doi:10.1016/j.aim.2025.110457](#).
13. 2025 **Number of singular points on projective surfaces.** **Jihao Liu**, Lingyao Xie. Chin. Ann. Math. Ser. B **46** (2025), 713–724. [arXiv:2103.04522](#), [doi:10.1007/s11401-025-0035-y](#).
14. 2024 **On effective Iitaka fibrations and existence of complements.** Guodu Chen, Jingjun Han, **Jihao Liu**. Int. Math. Res. Not. (2024), no. 10, 8329–8349. [arXiv:2301.04813](#), [doi:10.1093/imrn/rnad253](#).
15. 2024 **Uniform rational polytopes of foliated threefolds and the global ACC.** **Jihao Liu**, Fanjun Meng, Lingyao Xie. J. Lond. Math. Soc. **109** (2024), no. 6, e12950. [arXiv:2306.00330](#), [doi:10.1112/jlms.12950](#).
16. 2024 **Complements, index theorem, and minimal log discrepancies of foliated surface singularities.** **Jihao Liu**, Fanjun Meng, Lingyao Xie. Eur. J. Math. **10** (2024), no. 6. [arXiv:2305.06493](#), [doi:10.1007/s40879-023-00719-9](#).
17. 2024 **Infinitesimal structure of log canonical thresholds.** **Jihao Liu**, Fanjun Meng, Lingyao Xie. Doc. Math. **29** (2024), no. 3, 703–732. [arXiv:2209.11369](#), [doi:10.4171/dm/952](#).
18. 2024 **ACC for minimal log discrepancies of exceptional singularities.** Jingjun Han, **Jihao Liu**, Vyacheslav V. Shokurov. Peking Math. J. (2024), 1–33. [arXiv:1903.04338](#), [doi:10.1007/s42543-024-00091-x](#).
19. 2023 **Semi-ampleness of NQC generalized log canonical pairs.** **Jihao Liu**, Lingyao Xie. Adv. Math. **427** (2023), 109126. [arXiv:2210.01731](#), [doi:10.1016/j.aim.2023.109126](#).
20. 2023 **Existence of flips for generalized lc pairs.** Christopher D. Hacon, **Jihao Liu**. Camb. J. Math. **11** (2023), no. 4, 795–828. [arXiv:2105.13590](#), [doi:10.4310/CJM.2023.v11.n4.a1](#).
21. 2023 **On global ACC for foliated threefolds.** **Jihao Liu**, Yujie Luo, Fanjun Meng. Trans. Amer. Math. Soc. **376** (2023), no. 12, 8939–8972. [arXiv:2303.13083](#), [doi:10.1090/tran/9053](#).
22. 2023 **On the fixed part of pluricanonical systems for surfaces.** **Jihao Liu**, Lingyao Xie. Math. Nachr. **296** (2023), 2046–2069. [arXiv:2202.11260](#), [doi:10.1002/mana.202200088](#).
23. 2023 **Remark on complements on surfaces.** **Jihao Liu**. Forum Math. Sigma **11** (2023), e42. [arXiv:2208.09184](#), [doi:10.1017/fms.2023.35](#).

24. 2023 **Divisors computing minimal log discrepancies on lc surfaces.** Jihao Liu, Lingyao Xie. *Math. Proc. Camb. Philos. Soc.* **175** (2023), no. 1, 107–128. [arXiv:2101.00138](#), [doi:10.1017/S0305004123000051](#).
25. 2023 **On effective birationality for sub-pairs.** Jingjun Han, Jihao Liu. *Int. J. Math.* **34** (2023), no. 6, Article No. 2350029. [arXiv:2007.01849](#), [doi:10.1142/S0129167X23500295](#).
26. 2021 **An optimal gap of minimal log discrepancies of threefold non-canonical singularities.** Jihao Liu, Liudan Xiao. *J. Pure Appl. Algebra* **225** (2021), no. 9. [arXiv:1909.08759](#), [doi:10.1016/j.jpaa.2021.106674](#).
27. 2021 **Sarkisov program for generalized pairs.** Jihao Liu. *Osaka J. Math.* **58** (2021), no. 4. [arXiv:1802.03926](#), [doi:10.18910/84956](#).
28. 2020 **Bounded deformations of  $(\varepsilon, \delta)$ -log canonical singularities.** Jingjun Han, Jihao Liu, Joaquín Moraga. *J. Math. Sci. Univ. Tokyo* **27** (2020), no. 1, 1–28. [arXiv:1903.07202](#), pdf.

## PAPERS TO APPEAR

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Accepted but not yet published. Listed by arXiv posting date, most recent first.

1. 2024 **Volume of algebraically integrable foliations and locally stable families.** Jingjun Han, Junpeng Jiao, Mengchu Li, Jihao Liu. To appear in *Trans. Amer. Math. Soc.* [arXiv:2406.16604](#).
2. 2023 **The minimal volume of surfaces of log general type with non-empty non-klt locus.** Jihao Liu, Wenfei Liu. To appear in *Algebr. Geom.* [arXiv:2308.14268](#).
3. 2023 **ACC for lc thresholds for algebraically integrable foliations.** Omprokash Das, Jihao Liu, Roktim Mascharak. To appear in *Selecta Math.* [arXiv:2307.07157](#).
4. 2023 **Optimal bounds on surfaces.** Jihao Liu, Vyacheslav V. Shokurov. To appear in *Algebraic Geometry and Physics*. [arXiv:2305.19248](#).
5. 2018 **Toward the equivalence of the ACC for  $a$ -log canonical thresholds and the ACC for minimal log discrepancies.** Jihao Liu. To appear in *Algebraic Geometry and Physics*. [arXiv:1809.04839](#).

## PREPRINTS

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Listed by arXiv posting date, most recent first.

- 1a. 2026 **A Criteria of Weighted Homogeneity via Logarithmic Vector Fields.** Jihao Liu, Xiping Zhang. Not AI-generated; presents the same result as no. 1b below. [arXiv:2606.29886](#).
- 1b. 2026 **Criteria of isolated weighted homogeneous hypersurface singularities using Logarithmic vector fields.** Jihao Liu, Xiping Zhang. AI-generated (human-verified); presents the same result as no. 1a above. [arXiv:2606.29891](#).
2. 2026 **A counterexample to the near-quadratic Elekes–Rónyai expander conjecture over  $\mathbb{R}$ .** Jihao Liu. [arXiv:2606.16738](#).
3. 2026 **Examples of infinitely many  $(-1)$ -curves with higher genus on elliptic surfaces.** Sichen Li, Jihao Liu. [arXiv:2605.27926](#).
4. 2026 **Boundedness of total Cartier indices for rational singularities in families.** Jihao Liu, Ruicheng Hu, Sheng Qin. [arXiv:2605.22782](#).

5. 2026 **Shokurov's global index conjecture for threefold foliations.** Jihao Liu, Sheng Qin. [arXiv:2605.22735](#).
6. 2026 **A question on klt type varieties of Han and Jiang.** Jihao Liu. [arXiv:2605.22250](#).
7. 2026 **On a question of Mauri and Moraga.** Jihao Liu. [arXiv:2605.22052](#).
8. 2026 **Optimal bounds in bend-and-break for foliations.** Jihao Liu, Zeming Sun, Jiedong Jiang. [arXiv:2605.20754](#).
9. 2026 **An example of a very non-movable effective divisor.** Jihao Liu. [arXiv:2605.20594](#).
10. 2026 **On a question of Kollár and Kovács.** Jihao Liu. [arXiv:2605.20585](#).
11. 2026 **The minimal volume of stable surfaces of rank one.** Jihao Liu, Wenfei Liu. [arXiv:2605.05641](#).
12. 2026 **Birational boundedness of stable families.** Paolo Cascini, Jihao Liu, Calum Spicer, Roberto Svaldi. [arXiv:2604.24106](#).
13. 2026 **Existence of the minimal model program for log canonical generalized pairs.** Zhengyu Hu, Jihao Liu. [arXiv:2603.03817](#).
14. 2026 **Dual complexes of qdlt Fano type models and strong complete regularity.** Jihao Liu, Konstantin Loginov. [arXiv:2603.03588](#).
15. 2025 **Non-algebraicity of non-abundant foliations and abundance for adjoint foliated structures.** Jihao Liu, Zheng Xu. [arXiv:2510.04419](#).
16. 2025 **Variation of algebraically integrable adjoint foliated structures.** Paolo Cascini, Jihao Liu, Fanjun Meng, Roberto Svaldi, Lingyao Xie. [arXiv:2510.02498](#).
17. 2025 **Boundedness in general type MMP and fivefold effective termination / Effective termination of general type MMPs in dimension at most five.** Jingjun Han, Jihao Liu, Lu Qi (first paper only), Ziquan Zhuang. [arXiv:2506.20183v1](#) / [arXiv:2509.01501](#).
18. 2025 **Non-vanishing implies numerical dimension one abundance.** Jihao Liu, Zheng Xu. [arXiv:2505.05250](#).
19. 2025 **On finite generation and boundedness of adjoint foliated structures.** Paolo Cascini, Jingjun Han, Jihao Liu, Fanjun Meng, Calum Spicer, Roberto Svaldi, Lingyao Xie. [arXiv:2504.10737](#).
20. 2025 **Classification of threefold enc cDV quotient singularities.** Jingjun Han, Jihao Liu. [arXiv:2501.01024](#).
21. 2024 **ACC for local volumes.** Jingjun Han, Jihao Liu, Lu Qi. [arXiv:2408.15090](#).
22. 2024 **Minimal model program for algebraically integrable adjoint foliated structures.** Paolo Cascini, Jingjun Han, Jihao Liu, Fanjun Meng, Calum Spicer, Roberto Svaldi, Lingyao Xie. [arXiv:2408.14258](#).
23. 2023 **Minimal model program for algebraically integrable foliations and generalized pairs.** Guodu Chen, Jingjun Han, Jihao Liu, Lingyao Xie. [arXiv:2309.15823](#).
24. 2023 **Vanishing theorems for generalized pairs.** Bingyi Chen, Jihao Liu, Lingyao Xie. [arXiv:2305.12337](#).
25. 2022 **Second largest accumulation point of minimal log discrepancies of threefolds.** Jihao Liu, Yujie Luo. [arXiv:2207.04610](#).
26. 2018 **Accumulation point theorem for generalized log canonical thresholds.** Jihao Liu. [arXiv:1810.12381](#).
27. 2017 **On invariance of plurigenera for foliated surface pairs.** Jihao Liu. [arXiv:1707.07092](#).

## CONFERENCE AND WORKSHOP TALKS

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Selected; seminar talks omitted.

- Apr 2026 *Existence of the minimal model program for log canonical generalized pairs.*  
Algebraic Geometry Workshop 2026, YMSC, Tsinghua University.
- Feb 2026 *Boundedness of algebraically integrable foliations.*  
Algebraic Geometry Workshop, Sydney Mathematical Research Institute.
- Jan 2026 *Birational geometry of algebraically integrable adjoint foliated structures.*  
Workshop on Algebraic Geometry and Complex Geometry in Osaka, The University of Osaka.
- Dec 2025 *Boundedness of Fano algebraically integrable foliations.*  
Conference on Algebraic Geometry, TSIMF, Sanya.
- Dec 2025 *Boundedness of Fano algebraically integrable foliations* (online).  
Birational Geometry and Related Topics, XJTLU, Suzhou.
- Jun 2025 *Adjoint foliated structures in birational geometry.*  
Higher Dimensional Geometry and Foliations, Imperial College London.
- Jun 2025 *Abundance conjecture when  $\nu \leq 1$ .*  
Southeastern Algebraic Geometry Symposium VIII.
- Dec 2024 *Birational geometry of adjoint foliated structures.*  
Sanya Algebraic Geometry Workshop.
- Dec 2024 *Adjoint foliated structures on surfaces.*  
2024 代数几何研讨会, Tongji University.
- Nov 2024 *Some optimal bounds in explicit birational geometry.*  
2024 重庆理工大学代数几何会议, Chongqing University of Technology.
- Dec 2023 *Minimal model program for algebraically integrable foliations.*  
Sanya Algebraic Geometry Workshop.
- Jun 2023 *Multiplicities on lc places.*  
Birational Geometry Workshop, Tsinghua University.
- May 2023 *Optimal bounds for algebraic invariants of surfaces.*  
Simons Conference on Higher Dimensional Geometry, Simons Center.
- Aug 2022 *ACC for mlds for terminal threefolds.*  
Midwest Algebraic Geometry Graduate Conference, UIC.
- May 2022 *On Shokurov's ACC conjecture for mlds in dimension 3.*  
JAMI Conference 2022 (in honor of Shokurov's 70th birthday), Johns Hopkins University.
- Aug 2021 *Existence of flips for generalized pairs.*  
2021 Workshop on Algebraic Geometry: Generalised Pairs and Applications (online).
- Jul 2020 *Complements and ACC for minimal log discrepancies.*  
Simons Conference on Rationality (online).

## CONFERENCES AND SEMINARS ORGANIZED

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- Aug 2026 Conference on Foliation, Fibration, and Moduli.  
Westlake University, Hangzhou, August 24–29, 2026.  
With Chuanhao Wei (local organizer) and Paolo Cascini.

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| Jul 2026       | <b>5th National Algebraic Geometry Conference.</b><br>Peking University, July 6–10, 2026. Local organizer.<br>With Ruochuan Liu, Zhiyu Tian, Qizheng Yin.                                        |
| Aug 2025       | <b>Summer School on Birational Geometry of Foliations.</b><br>Peking University, August 18–22, 2025.<br>With Paolo Cascini, Jingjun Han, Fanjun Meng, Calum Spicer, Roberto Svaldi, Lingyao Xie. |
| May 2025       | <b>Conference on Explicit Birational Geometry.</b><br>Peking University, May 28 – June 1, 2025.                                                                                                  |
| May 2025       | <b>6th Beijing Algebraic Geometry Day.</b><br>Peking University, May 9, 2025. Local organizer.                                                                                                   |
| Nov 2024       | <b>2nd Beijing Algebraic Geometry Day.</b><br>Peking University, November 23, 2024. Local organizer.                                                                                             |
| Aug 2024       | <b>1st Conference on Foliation and Birational Geometry.</b><br>Fudan University, August 26–30, 2024.<br>With Meng Chen, Chen Jiang, Jingjun Han.                                                 |
| 2024 – present | <b>Peking University Algebraic Geometry Seminar.</b><br>With Fumiaki Suzuki (from 2025), Zhiyu Tian, Qizheng Yin.                                                                                |
| 2024 – present | Beijing Algebraic Geometry Day (joint workshop series with the Chinese Academy of Sciences, Tsinghua University, and Capital Normal University).                                                 |
| 2021 – 2024    | <b>Northwestern Algebraic Geometry Seminar.</b><br>Co-organized over six semesters with Yuchen Liu and rotating co-organizers (Charles Stibitz, Kai Huang, Ananth Shankar).                      |
| 2019 – 2020    | <b>Utah Algebraic Geometry Reading Seminar.</b><br>University of Utah, Fall 2019 – Spring 2020.                                                                                                  |

## TEACHING

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### PEKING UNIVERSITY

|             |                                                     |
|-------------|-----------------------------------------------------|
| Spring 2026 | 代数几何 II (Algebraic Geometry II) — instructor.       |
| Fall 2025   | 代数几何 I (Algebraic Geometry I) — instructor.         |
| Spring 2025 | 代数几何专题 (Topics in Algebraic Geometry) — instructor. |

### NORTHWESTERN UNIVERSITY

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|-------------|------------------------------------------------------------------------------------------------------------------------|
| Winter 2024 | Math 230-2 — Multivariable Integral Calculus. Instructor and course coordinator (sections 31, 41).                     |
| Fall 2023   | Math 220-1 — Single-Variable Differential Calculus. Instructor (sections 71, 81).                                      |
| Winter 2023 | Math 220-2 — Single-Variable Integral Calculus. Instructor (sections 31, 41).                                          |
| Fall 2022   | Math 228-1 — Multivariable Differential Calculus for Engineering. Instructor (sections 71, 81).                        |
| Winter 2022 | Math 228-1 — Multivariable Differential Calculus for Engineering. Instructor and course coordinator (sections 41, 51). |
| Fall 2021   | Math 220-1 — Single-Variable Differential Calculus. Instructor (sections 41, 53).                                      |

## UNIVERSITY OF UTAH

2017 – 2021

Math 1090 (Business Algebra) — instructor, Spring 2021, Fall 2020, Fall 2019.

Math 2250 (Differential Equations) — teaching assistant, Fall 2018, Spring 2018.

Math 1320 (Engineering Calculus) — teaching assistant, Spring 2017.

## REFEREEING

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Refereed for the following journals (in alphabetical order):

Acta Mathematica Sinica (English Series) · Advances in Mathematics · Algebra and Number Theory · Algebraic Geometry and Physics · Annals of Mathematics · Communications in Analysis and Geometry · Compositio Mathematica · Electronic Research Archive · Épijournal de Géométrie Algébrique · European Journal of Mathematics · Forum of Mathematics, Sigma · International Journal of Mathematics · International Mathematics Research Notices · Journal für die reine und angewandte Mathematik (Crelle's Journal) · Journal of Algebra · Journal of Algebraic Geometry · Journal of Differential Geometry · Journal of the European Mathematical Society · Journal of the Institute of Mathematics of Jussieu · Journal of the London Mathematical Society · Manuscripta Mathematica · Mathematische Annalen · Michigan Mathematical Journal · Pacific Journal of Mathematics · Peking Mathematical Journal · Proceedings of the American Mathematical Society · Proceedings of the London Mathematical Society · Pure and Applied Mathematics Quarterly · Selecta Mathematica · Surveys in Differential Geometry.

Also refereed for *Higher Dimensional Algebraic Geometry: A Volume in Honor of V. V. Shokurov* (London Math. Soc. Lecture Note Ser.). Formerly a reviewer for Mathematical Reviews.